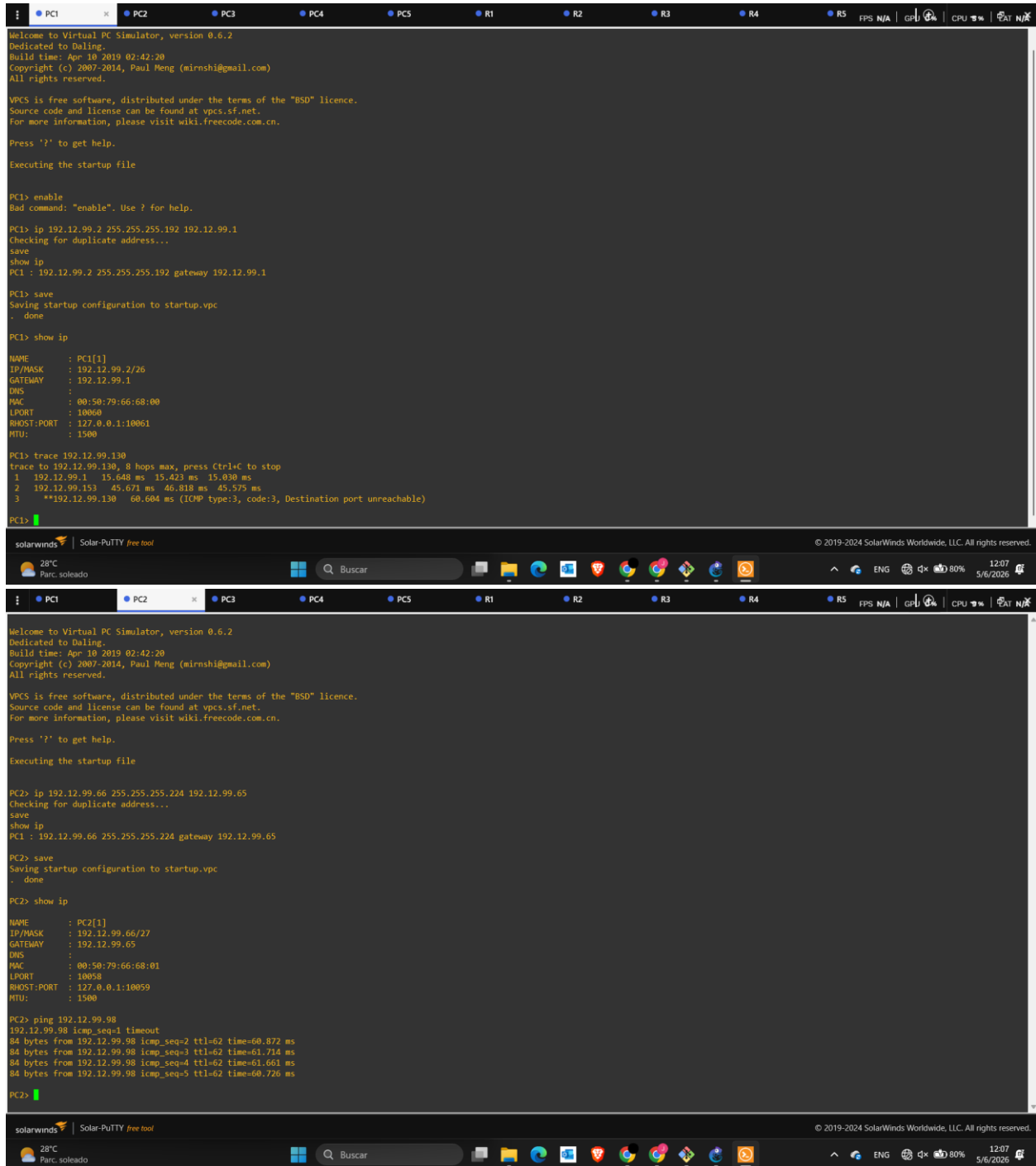


Ejercicio 2 parcial 02- RDC

Carnet: 00018824; A: 12; B: 99; Julio Alejandro Flores Diaz

Red: 192.12.99.0/24



```
Welcome to Virtual PC Simulator, version 0.6.2
Dedicated to Daling.
Build time: Apr 10 2019 02:42:20
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For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC1> enable
Bad command: "enable". Use ? for help.

PC1> ip 192.12.99.2 255.255.255.192 192.12.99.1
Checking for duplicate address...
save
show ip
PC1 : 192.12.99.2 255.255.255.192 gateway 192.12.99.1

PC1> save
Saving startup configuration to startup.vpc
. done

PC1> show ip

NAME       : PC1[1]
IP/MASK    : 192.12.99.2/26
GATEWAY    : 192.12.99.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 10060
RHOST:PORT : 127.0.0.1:10061
MTU        : 1500

PC1> trace 192.12.99.130
Trace to 192.12.99.130, 8 hops max, press Ctrl+C to stop
 1  192.12.99.1   15.648 ms  15.423 ms  15.030 ms
 2  192.12.99.153 45.671 ms  46.818 ms  45.575 ms
 3  **192.12.99.130 60.604 ms (ICMP type:3, code:3, Destination port unreachable)

PC1>
```

```
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Press '?' to get help.

Executing the startup file

PC2> ip 192.12.99.66 255.255.255.224 192.12.99.65
Checking for duplicate address...
save
show ip
PC1 : 192.12.99.66 255.255.255.224 gateway 192.12.99.65

PC2> save
Saving startup configuration to startup.vpc
. done

PC2> show ip

NAME       : PC2[1]
IP/MASK    : 192.12.99.66/27
GATEWAY    : 192.12.99.65
DNS        :
MAC        : 00:50:79:66:68:01
LPORT      : 10059
RHOST:PORT : 127.0.0.1:10059
MTU        : 1500

PC2> ping 192.12.99.98
192.12.99.98 icmp_seq=1 timeout
84 bytes from 192.12.99.98 icmp_seq=2 ttl=62 time=60.872 ms
84 bytes from 192.12.99.98 icmp_seq=3 ttl=62 time=61.714 ms
84 bytes from 192.12.99.98 icmp_seq=4 ttl=62 time=61.661 ms
84 bytes from 192.12.99.98 icmp_seq=5 ttl=62 time=60.726 ms

PC2>
```

PC1PC2PC3PC4PC5R1R2R3R4R5FPS N/A | GPU 0% | CPU 7% | NAT N/A

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Press '?' to get help.

Executing the startup file

PC3> ip 192.12.99.98 255.255.255.240 192.12.99.97
Checking for duplicate address...
save
show PC1 : 192.12.99.98 255.255.255.240 gateway 192.12.99.97

PC3> save
Saving startup configuration to startup.vpc
.: done

PC3> show ip

NAME	: PC3[1]
IP/MASK	: 192.12.99.98/28
GATEWAY	: 192.12.99.97
DNS	:
MAC	: 00:50:79:66:68:02
LPORT	: 10056
RHOST:PORT	: 127.0.0.1:10057
MTU:	: 1500

PC3> trace 192.12.99.114
trace to 192.12.99.114, 8 hops max, press Ctrl+C to stop
1 192.12.99.97 15.654 ms 15.186 ms 16.198 ms
2 192.12.99.141 45.698 ms 45.357 ms 46.078 ms
3 **192.12.99.114 60.815 ms (ICMP type:3, code:3, Destination port unreachable)

PC3> █

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Buscar

ENG 80% 12:08
5/6/2026

PC1PC2PC3PC4PC5R1R2R3R4R5FPS N/A | GPU 0% | CPU 7% | NAT N/A

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Press '?' to get help.

Executing the startup file

PC4> ip 192.12.99.114 255.255.255.240 192.12.99.113
Checking for duplicate address...
save
show PC1 : 192.12.99.114 255.255.255.240 gateway 192.12.99.113

PC4> save
Saving startup configuration to startup.vpc
.: done

PC4> show ip

NAME	: PC4[1]
IP/MASK	: 192.12.99.114/28
GATEWAY	: 192.12.99.113
DNS	:
MAC	: 00:50:79:66:68:03
LPORT	: 10062
RHOST:PORT	: 127.0.0.1:10063
MTU:	: 1500

PC4> ping 192.12.99.130
84 bytes from 192.12.99.130 icmp_seq=1 ttl=61 time=92.224 ms
84 bytes from 192.12.99.130 icmp_seq=2 ttl=61 time=91.294 ms
84 bytes from 192.12.99.130 icmp_seq=3 ttl=61 time=90.637 ms
84 bytes from 192.12.99.130 icmp_seq=4 ttl=61 time=91.008 ms
84 bytes from 192.12.99.130 icmp_seq=5 ttl=61 time=90.521 ms

PC4> █

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Buscar

ENG 80% 12:08
5/6/2026

PC1PC2PC3PC4PC5R1R2R3R4R5FPS N/A GPU CPU 80% NAT

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Press '?' to get help.

Executing the startup file

PC5> ip 192.12.99.130 255.255.255.248 192.12.99.129
Checking for duplicate address...
save
show ip
PC1 : 192.12.99.130 255.255.255.248 gateway 192.12.99.129

PC5> save
Saving startup configuration to startup.vpc
- done

PC5> show ip

NAME : PC5[1]
IP/MASK : 192.12.99.130/20
GATEWAY : 192.12.99.129
DNS :
MAC : 00:50:79:66:68:04
LPORT : 10064
RHOST:PORT : 127.0.0.1:10065
MTU : 1500

PC5> trace 192.12.99.66
trace to 192.12.99.66, 8 hops max, press Ctrl+C to stop
1 192.12.99.129 15.729 ms 15.419 ms 15.197 ms
2 192.12.99.154 47.227 ms 45.617 ms 45.252 ms
3 192.12.99.157 76.412 ms 76.094 ms 75.541 ms
4 *192.12.99.66 90.537 ms (ICMP type:3, code:3, Destination port unreachable)

PC5>

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PC1PC2PC3PC4PC5R1R2R3R4R5FPS N/A GPU CPU 80% NAT

*Mar 1 00:16:15.711: %OSPF-5-ADJCHG: Process 1, Nbr 192.12.99.149 on Serial2/1 from LOADING to FULL, Loading Done
R1#network 192.12.99.136 0.0.0.3 area 0
^
% Invalid input detected at '^' marker.

R1#
Building configuration...
[OK]
R1#show ip inter
R1#show ip interface br
R1#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.12.99.113	YES	manual	up	up
FastEthernet0/1	unassigned	YES	unset	administratively down	down
FastEthernet1/0	unassigned	YES	unset	administratively down	down
Serial1/0	192.12.99.137	YES	manual	up	up
Serial2/1	192.12.99.141	YES	manual	up	up
Serial2/2	unassigned	YES	unset	administratively down	down
Serial2/3	unassigned	YES	unset	administratively down	down

R1#show ip rou
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODM, P - periodic downloaded static route

Gateway of last resort is not set

192.12.99.0/24 is variably subnetted, 11 subnets, 5 masks
C 192.12.99.140/30 is directly connected, Serial2/1
C 192.12.99.136/30 is directly connected, Serial2/0
O 192.12.99.128/29 [110/75] via 192.12.99.142, 00:02:23, Serial2/1
O 192.12.99.156/30 [110/65] via 192.12.99.138, 00:05:22, Serial2/0
O 192.12.99.152/30 [110/75] via 192.12.99.142, 00:02:33, Serial2/1
O 192.12.99.148/30 [110/75] via 192.12.99.138, 00:00:55, Serial2/0
O 192.12.99.148/30 [110/65] via 192.12.99.142, 00:04:16, Serial2/1
O 192.12.99.144/30 [110/74] via 192.12.99.142, 00:04:26, Serial2/1
O 192.12.99.144/30 [110/74] via 192.12.99.138, 00:06:03, Serial2/0
O 192.12.99.0/26 [110/75] via 192.12.99.138, 00:00:57, Serial2/0
O 192.12.99.96/28 [110/74] via 192.12.99.142, 00:03:53, Serial2/1
C 192.12.99.112/28 is directly connected, FastEthernet0/0
O 192.12.99.64/27 [110/74] via 192.12.99.138, 00:08:40, Serial2/0

R1#

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PC1

PC2

PC3

PC4

PC5

R1

R2

R3

R4

R5

FPS N/A

CPU 11%

GPU 100%

RAM 100%

R2#

Building configuration...

[OK]

R2#

*Mar 1 00:14:55.391: %SYS-5-CONFIG_I: Configured from console by console

R2#

*Mar 1 00:15:49.595: %OSPF-5-ADJCHG: Process 1, Nbr 192.12.99.149 on FastEthernet0/1 from LOADING to FULL, Loading Done

R2#

*Mar 1 00:19:11.163: %OSPF-5-ADJCHG: Process 1, Nbr 192.12.99.158 on FastEthernet1/0 from LOADING to FULL, Loading Done

R2#show ip inter

R2#show ip interface br

R2#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.12.99.65	YES	manual	up	up
FastEthernet0/1	192.12.99.145	YES	manual	up	up
FastEthernet1/0	192.12.99.157	YES	manual	up	up
Serial2/0	192.12.99.138	YES	manual	up	up
Serial2/1	unassigned	YES	unset	administratively down	down
Serial2/2	unassigned	YES	unset	administratively down	down
Serial2/3	unassigned	YES	unset	administratively down	down

R2#show ip rou

R2#show ip route

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - OOR, P - periodic downloaded static route

Gateway of last resort is not set

192.12.99.0/24 is variably subnetted, 11 subnets, 5 masks

O 192.12.99.140/30 [110/74] via 192.12.99.146, 00:04:08, FastEthernet0/1

C 192.12.99.136/30 is directly connected, Serial2/0

O 192.12.99.128/29 [110/21] via 192.12.99.150, 00:00:39, FastEthernet1/0

C 192.12.99.120/24 [110/21] via 192.12.99.146, 00:02:07, FastEthernet0/1

C 192.12.99.156/30 is directly connected, FastEthernet1/0

O 192.12.99.152/30 [110/11] via 192.12.99.158, 00:00:39, FastEthernet1/0

O 192.12.99.148/30 [110/11] via 192.12.99.146, 00:04:00, FastEthernet0/1

C 192.12.99.144/30 is directly connected, FastEthernet0/1

O 192.12.99.0/26 [110/11] via 192.12.99.150, 00:00:51, FastEthernet1/0

O 192.12.99.96/28 [110/20] via 192.12.99.146, 00:03:37, FastEthernet0/1

O 192.12.99.112/28 [110/74] via 192.12.99.137, 00:06:03, Serial2/0

C 192.12.99.64/27 is directly connected, FastEthernet0/0

R2#

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28°C

Parc soleado

ENG

80%

12:08

5/6/2026

PC1

PC2

PC3

PC4

PC5

R1

R2

R3

R4

R5

FPS N/A

CPU 11%

GPU 100%

RAM 100%

R3(config-router)#network 192.12.99.96 0.0.0.15 area 0

R3(config-router)#end

R3#

Building configuration...

[OK]

R3#

*Mar 1 00:15:00.259: %SYS-5-CONFIG_I: Configured from console by console

R3#

*Mar 1 00:16:12.767: %OSPF-5-ADJCHG: Process 1, Nbr 192.12.99.153 on FastEthernet1/0 from LOADING to FULL, Loading Done

R3#show ip inter

R3#show ip interface br

R3#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.12.99.97	YES	manual	up	up
FastEthernet0/1	192.12.99.146	YES	manual	up	up
FastEthernet1/0	192.12.99.149	YES	manual	up	up
Serial2/0	unassigned	YES	unset	administratively down	down
Serial2/1	192.12.99.142	YES	manual	up	up
Serial2/2	unassigned	YES	unset	administratively down	down
Serial2/3	unassigned	YES	unset	administratively down	down

R3#show ip rou

R3#show ip route

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - OOR, P - periodic downloaded static route

Gateway of last resort is not set

192.12.99.0/24 is variably subnetted, 11 subnets, 5 masks

C 192.12.99.140/30 is directly connected, Serial2/1

O 192.12.99.136/30 [110/74] via 192.12.99.145, 00:04:00, FastEthernet0/1

O 192.12.99.128/29 [110/11] via 192.12.99.150, 00:01:58, FastEthernet1/0

O 192.12.99.156/30 [110/11] via 192.12.99.145, 00:04:00, FastEthernet0/1

O 192.12.99.152/30 [110/11] via 192.12.99.150, 00:02:00, FastEthernet1/0

C 192.12.99.148/30 is directly connected, FastEthernet1/0

C 192.12.99.144/30 is directly connected, FastEthernet0/1

O 192.12.99.0/26 [110/21] via 192.12.99.150, 00:00:32, FastEthernet1/0

C 192.12.99.0/26 [110/21] via 192.12.99.145, 00:00:32, FastEthernet0/1

O 192.12.99.96/28 is directly connected, FastEthernet0/0

O 192.12.99.112/28 [110/74] via 192.12.99.141, 00:04:27, Serial2/1

O 192.12.99.64/27 [110/20] via 192.12.99.145, 00:04:01, FastEthernet0/1

R3#

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28°C

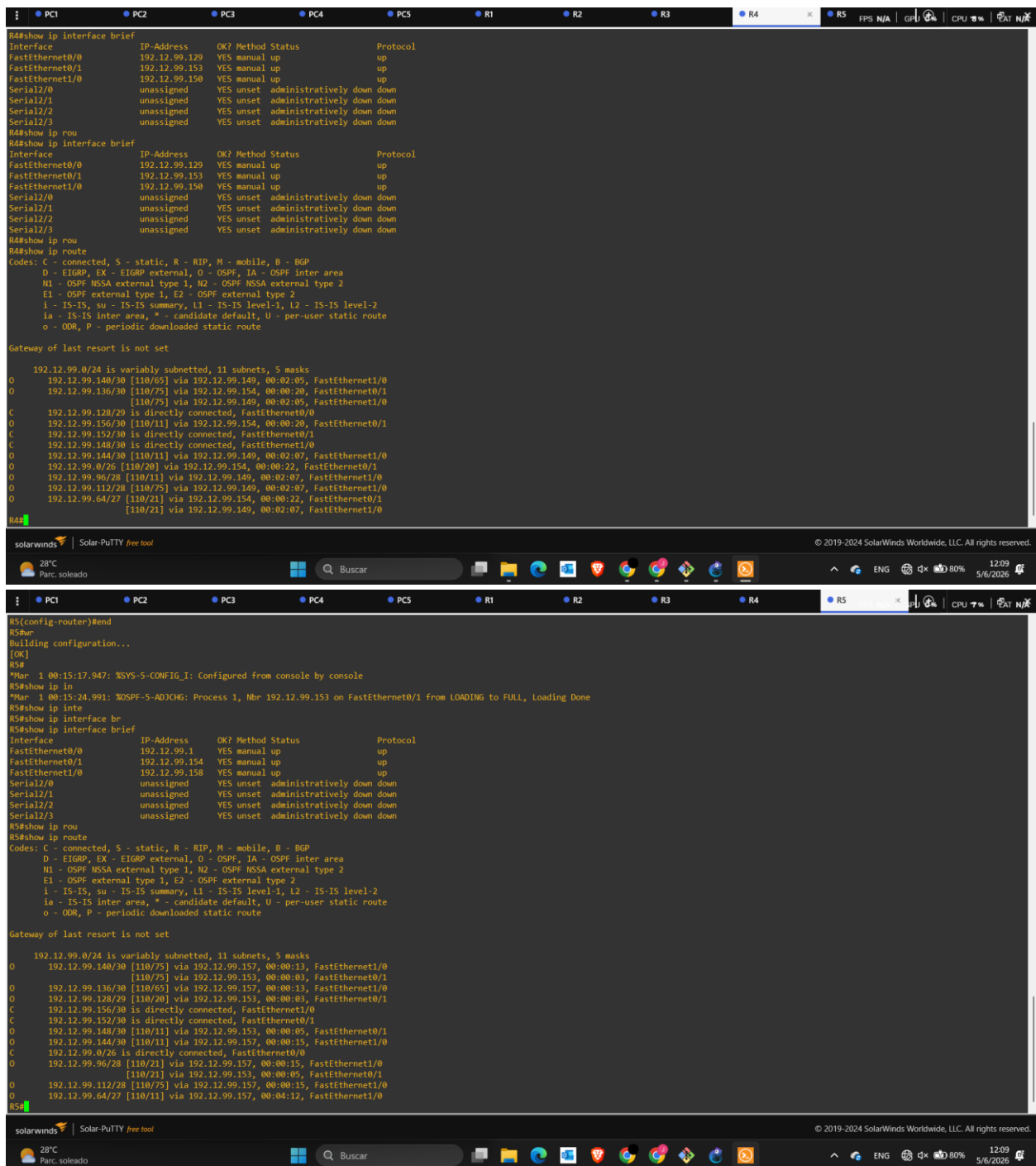
Parc soleado

ENG

80%

12:09

5/6/2026



wild mask

0.0.0.15

0.0.0.3

0.0.0.3

0.0.0.3

0.0.0.3

0.0.0.3

0.0.0.3

0.0.0.3

0.0.0.15

0.0.0.3

0.0.0.3

0.7

0.3

0.3

63

63

31

15

15

7

Red: 192.12.99.0/24 → 32-27 = hosts

Pc1: 50

Pc2: 25

Pc3: 12

Pc4: 8

Pc5: 6

R1-R2: 2

R3-R4: 2

R4-R5: 2

R5-R2: 2

Pc1: 2⁶ → 64

32-6 = 126

126 = ... 192

Pc4: 2⁴ → 16

32-4 = 128

128 = ... 240

Pc1

Pc2

Pc3

Pc4

Pc5

R1-R2

R1-R3

R3-R4

R4-R5

R5-R2

Pc2: 2⁵ → 32

32-5 = 127

127 = ... 224

Pc5: 2³ → 8

32-3 = 129

129 = ... 248

Ran 90

192.12.99.62

192.12.99.64

192.12.99.65

192.12.99.91

192.12.99.96

192.12.99.97

192.12.99.110

192.12.99.112

192.12.99.113

192.12.99.128

192.12.99.129

192.12.99.134

192.12.99.136

192.12.99.137

192.12.99.140

192.12.99.141

192.12.99.142

192.12.99.145

192.12.99.146

192.12.99.149

192.12.99.150

Pc3: 2⁴ → 16

32-4 = 128

128 = ... 240

R1-R2: 2² → 4

32-2 = 130

130 = ... 252

mask

192.12.99.126

192.12.99.127

192.12.99.128

192.12.99.129

192.12.99.130

192.12.99.131

192.12.99.132

192.12.99.133

192.12.99.134

192.12.99.135

192.12.99.136

192.12.99.137

192.12.99.138

192.12.99.139

192.12.99.140

192.12.99.141

192.12.99.142

192.12.99.143

192.12.99.144

192.12.99.145

192.12.99.146

192.12.99.147

192.12.99.148

192.12.99.149

192.12.99.150

R1-R2

R3-R4

R5-R2

R1-R3

R3-R4

R4-R5

R5-R2

R1-R2

R3-R4

R4-R5

R5-R2

R1-R3

R3-R4

R4-R5

R5-R2

R1-R2

R3-R4

R4-R5

R5-R2

R1-R3

R3-R4

R4-R5

R5-R2

R1-R2

R3-R4

R4-R5

R5-R2

R1-R3

R3-R4

R4-R5

R5-R2

R1-R2

R3-R4

R4-R5

R5-R2

126=...192 } 127=...224 } 128=...240

129=...248 } 130=...252

DISP	Interface	IP	mask	gateway	wildmask
✓ R1	Fe 0/0 - P4	192.12.99.113	128	NA	0.0.0.15
✓ R1	Serial 2/0 - R2	192.12.99.137	130	NA	0.0.0.3
✓ R1	Serial 2/1 - R3	192.12.99.141	130	NA	0.0.0.3
✓ R2	Serial 2/0 - R1	192.12.99.138	130	NA	0.0.0.3
✓ R2	Fe 0/0 - P2	192.12.99.65	127	NA	0.0.0.31
✓ R2	Fe 0/1 - R3	192.12.99.145	130	NA	0.0.0.3
✓ R2	Fe 1/0 - R5	192.12.99.157	130	NA	0.0.0.3
R3	Serial 2/1 - R1	192.12.99.142	130	NA	0.0.0.3
R3	Fe 0/0 - P3	192.12.99.97	128	NA	0.0.0.15
R3	Fe 0/1 - R2	192.12.99.146	130	NA	0.0.0.3
R3	Fe 1/0 - R4	192.12.99.149	130	NA	0.0.0.3
R4	Fe 0/0 - P5	192.12.99.129	129	NA	0.0.0.7
R4	Fe 0/1 - R5	192.12.99.153	130	NA	0.0.0.3
R4	Fe 1/0 - R3	192.12.99.150	130	NA	0.0.0.3
R5	Fe 0/0 - P1	192.12.99.1	126	NA	0.0.0.63
R5	Fe 0/1 - R4	192.12.99.154	130	NA	0.0.0.3
R5	Fe 1/0 - R2	192.12.99.158	130	NA	0.0.0.3
P01	Holo - R5	192.12.99.2	126	192.12.99.1	0.0.0.63
P02	Holo - R2	192.12.99.66	127	192.12.99.65	0.0.0.31
P03	Holo - R3	192.12.99.98	128	192.12.99.97	0.0.0.15
P04	Holo - R1	192.12.99.114	128	192.12.99.113	0.0.0.15
P05	Holo - R4	192.12.99.130	129	192.12.99.129	0.0.0.7